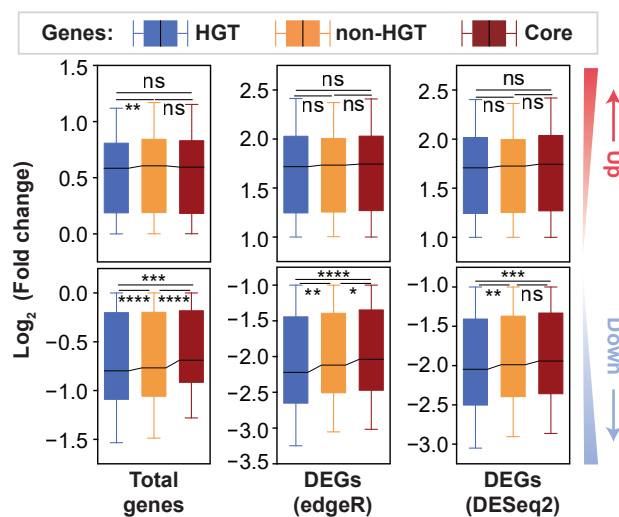




The diagram illustrates the classification of genes into three categories based on their evolutionary origin and function:

- HGT genes** (Horizontal Gene Transfer genes): Represented by a dashed box containing numerous small circles, indicating a large pool of genes acquired from other species.
- Complementary genes (C-genes)**: Indicated by a downward arrow from HGT genes, these are genes that complement the host's genome.
- Redundant genes (R-genes)**: Represented by two circular genomes (light blue) with internal colored segments (grey, yellow, green, blue, orange). A pink arrow points to a specific segment in both genomes, labeled as **Redundant genes (R-genes)**, indicating genes that are duplicated or have similar functions to existing host genes.

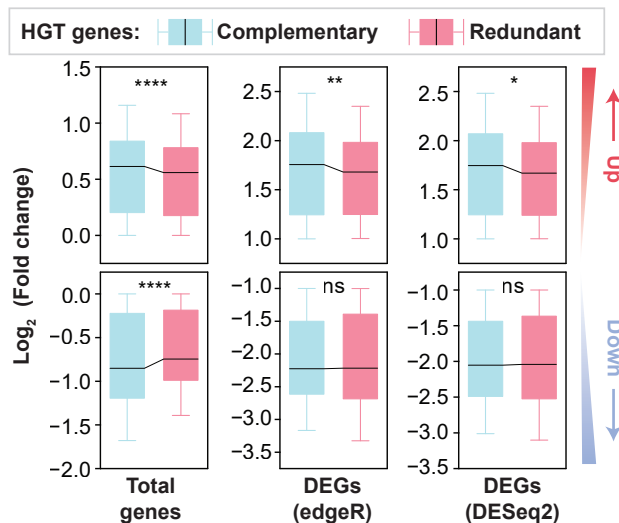
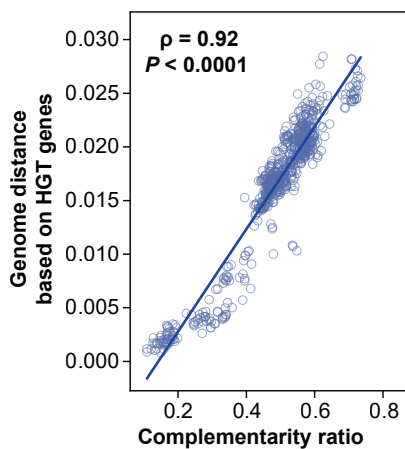


Diagram illustrating the calculation of the Complementarity ratio:

$$\text{Complementarity ratio} = \frac{\text{Counts of C-genes}}{\text{Counts of Total genes}}$$

The diagram shows two circular genomes, one with a blue nucleus and one with a red nucleus, each divided into colored segments representing genes. The top genome is labeled "Total genes" and the bottom one "Complementarity ratio". The bottom genome is also labeled "HGT genes".



$\rho = 0.34$
 $P < 0.0001$

Relative fitness

Complementarity

y ratio

> 0.5 < 0.5

Abundance & Counts

Abundance of up-regulated DEGs

- × 0
- 0.3
- 0.5
- 0.7

Average Fold change of DEGs

- 1
- 2
- 4

Fold change ▶ Average fold change of core DEGs - - - Fold change = 1

Gene	Core DEGs (Avg Fold Change)	Non-core Set (Fold Change)
J	1.09	1.58, 1.71
K	1.17	1.55, 1.77
L	0.34	2.39, 1.90
C	3.02	4.88, 1.75
E	2.82	2.05, 1.86
F	1.40	1.65, 1.35
G	1.44	1.75, 1.67
H	1.41	1.54, 1.77
I	3.65	3.97, 2.95
P	1.91	1.82, 1.89
Q	0.25	2.85, 2.03
D	1.94	1.12, 1.85
M	2.02	1.96, 1.99
N	ns	0.15, 0.24
O	1.67	2.33, 2.38
T	1.23	2.27, 2.46
U	1.51	1.26, 1.42
V	0.82	1.07, 1.76